

Intro to ARMA models

FISH 550 – Applied Time Series Analysis

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Topics for today

Review

- White noise
- Random walks

Autoregressive (AR) models

Moving average (MA) models

Autoregressive moving average (ARMA) models

Using ACF & PACF for model ID

Code for today

You can find the R code for these lecture notes and other related exercises [here](#).

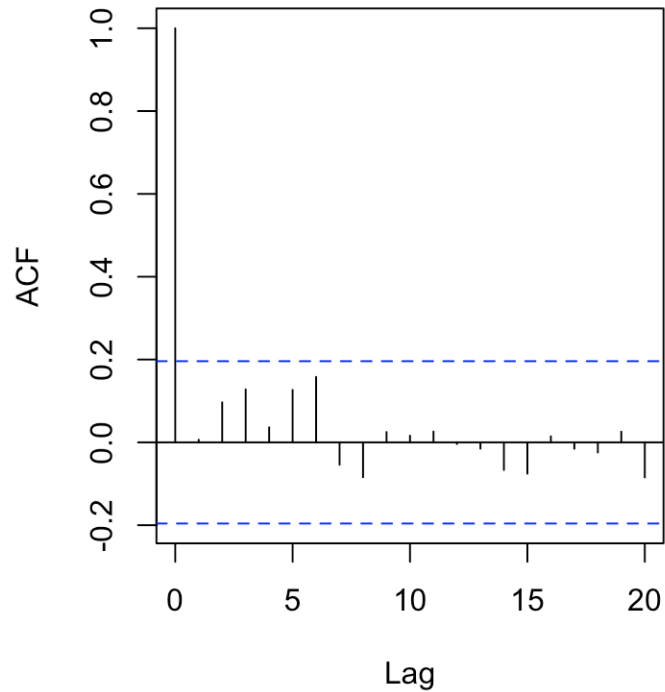
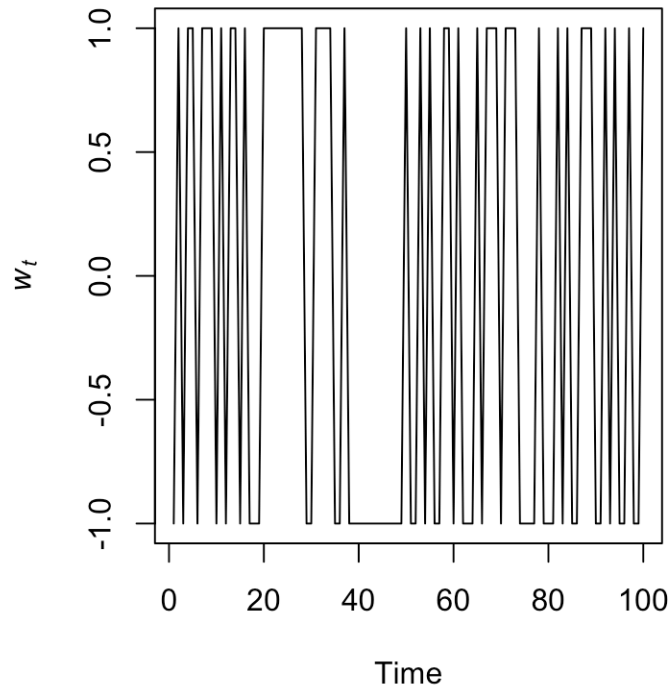
White noise (WN)

A time series $\{w_t\}$ is discrete white noise if its values are

1. independent
2. identically distributed with a mean of zero

The distributional form for $\{w_t\}$ is flexible

White noise (WN)



$$w_t = 2e_t - 1; e_t \sim \text{Bernoulli}(0.5)$$

Gaussian white noise

We often assume so-called *Gaussian white noise*, whereby

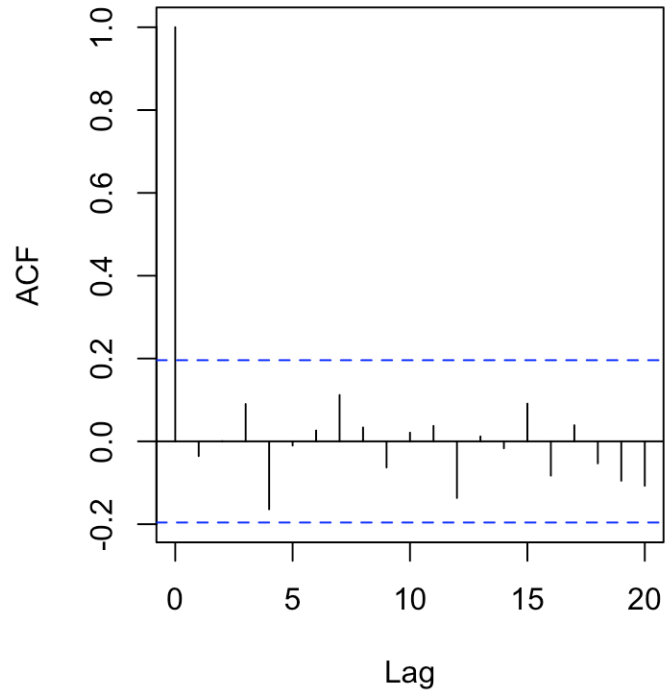
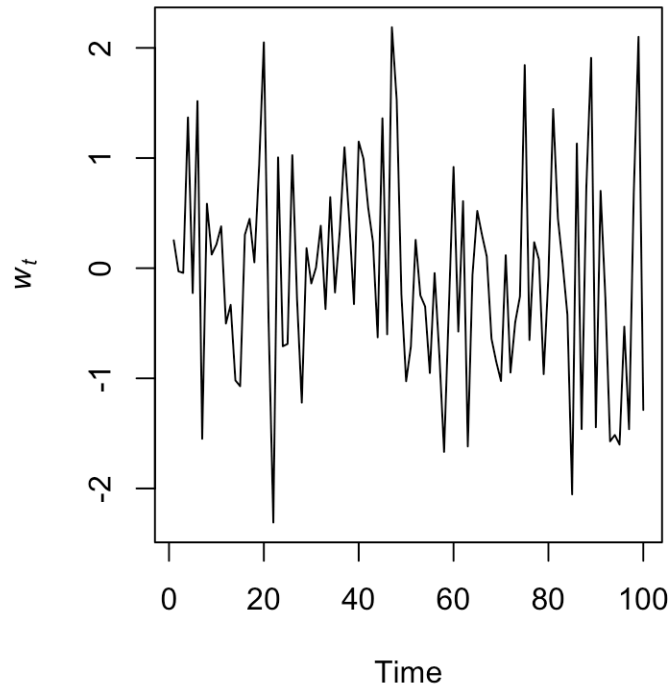
$$w_t \sim \mathcal{N}(0, \sigma^2)$$

and the following apply as well

$$\text{autocovariance: } \gamma_k = \begin{cases} \sigma^2 & \text{if } k = 0 \\ 0 & \text{if } k \geq 1 \end{cases}$$

$$\text{autocorrelation: } \rho_k = \begin{cases} 1 & \text{if } k = 0 \\ 0 & \text{if } k \geq 1 \end{cases}$$

Gaussian white noise



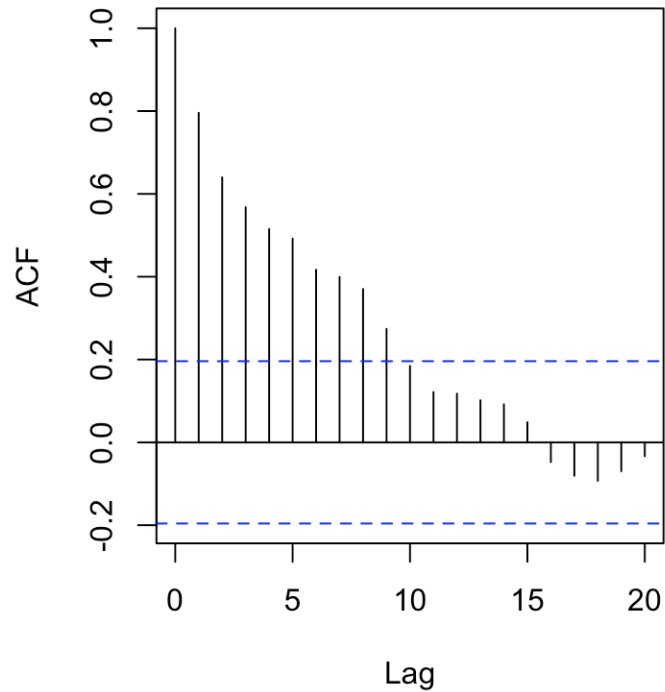
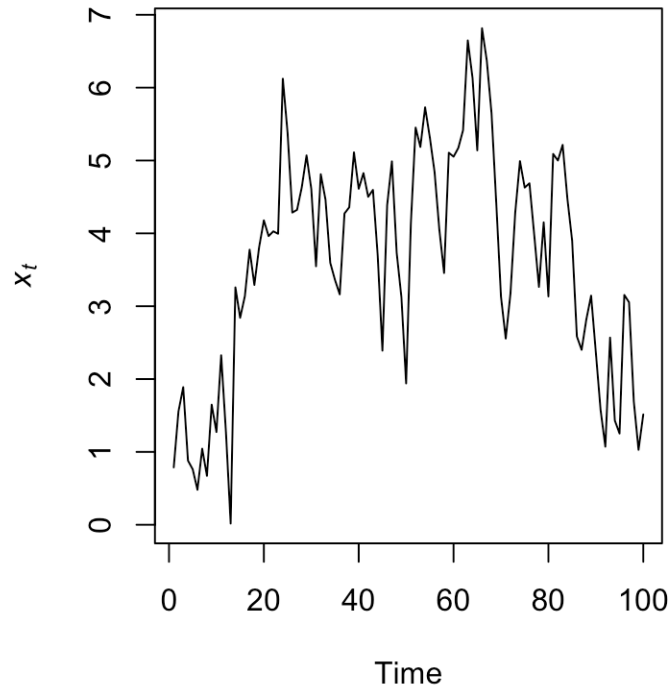
$$w_t \sim N(0, 1)$$

Random walk (RW)

A time series $\{x_t\}$ is a random walk if

1. $x_t = x_{t-1} + w_t$
2. w_t is white noise

Random walk (RW)



$$x_t = x_{t-1} + w_t; w_t \sim N(0, 1)$$

Random walk (RW)

Of note: Random walks are extremely flexible models and can be fit to many kinds of time series

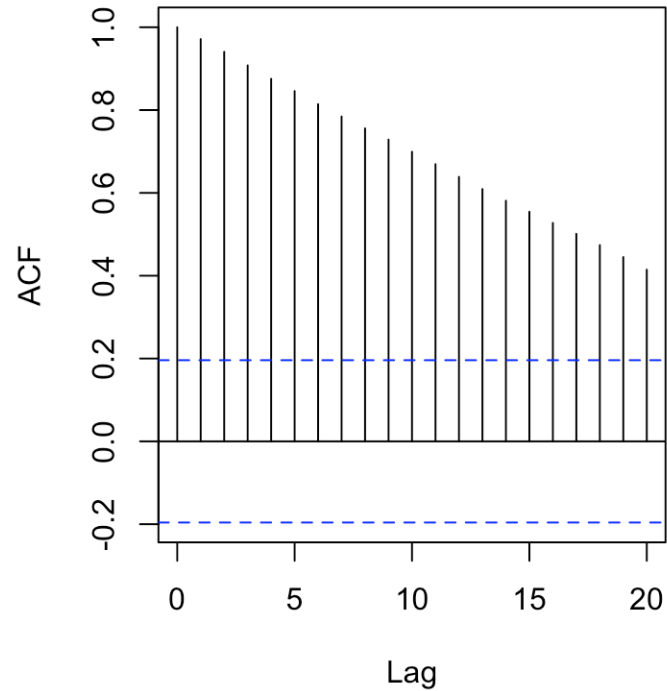
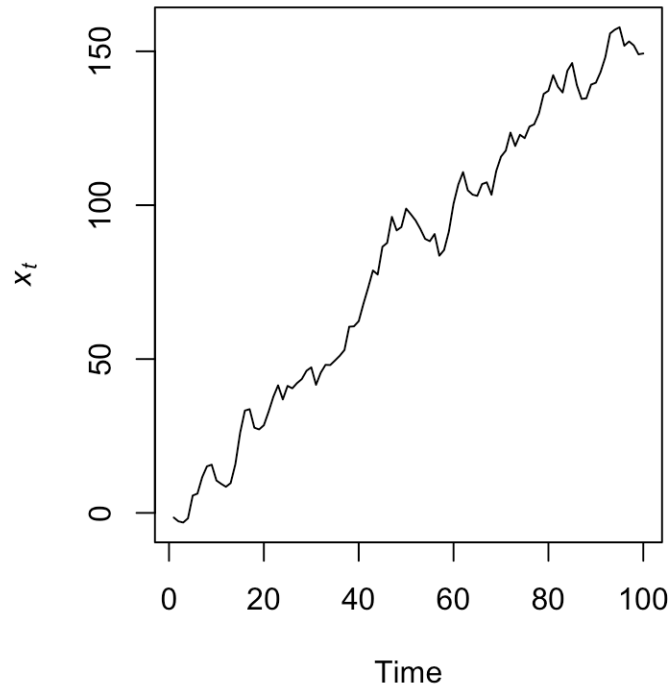
Biased random walk

A *biased random walk* (or *random walk with drift*) is written as

$$x_t = x_{t-1} + u + w_t$$

where u is the bias (drift) per time step and w_t is white noise

Biased random walk



$$x_t = x_{t-1} + 1 + w_t; w_t \sim N(0, 4)$$

Differencing a biased random walk

First-differencing a biased random walk yields a constant mean (level) u plus white noise

$$x_t = x_{t-1} + u + w_t$$

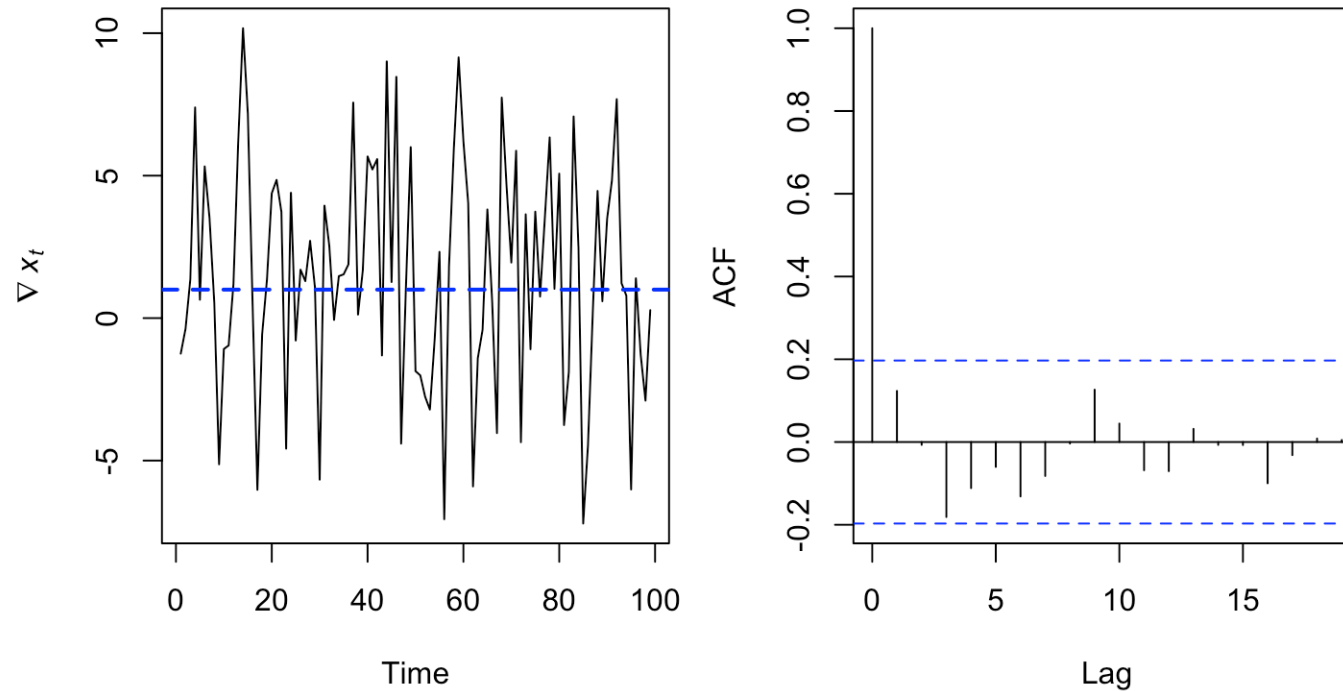
$$\Downarrow$$

$$\nabla(x_t = x_{t-1} + u + w_t)$$

$$x_t - x_{t-1} = x_{t-1} + u + w_t - x_{t-1}$$

$$x_t - x_{t-1} = u + w_t$$

Differencing a biased random walk



$$x_t - x_{t-1} = 1 + w_t; w_t \sim N(0, 1)$$

Linear stationary models

Linear stationary models

We saw last week that linear filters are a useful way of modeling time series

Here we extend those ideas to a general class of models call *autoregressive moving average* (ARMA) models

Autoregressive (AR) models

Autoregressive models are widely used in ecology to treat a current state of nature as a function its past state(s)

Autoregressive (AR) models

An *autoregressive* model of order p , or $AR(p)$, is defined as

$$x_t = \phi_1 x_{t-1} + \phi_2 x_{t-2} + \cdots + \phi_p x_{t-p} + w_t$$

where we assume

1. w_t is white noise
2. $\phi_p \neq 0$ for an order- p process

Examples of AR(p) models

AR(1)

$$x_t = 0.5x_{t-1} + w_t$$

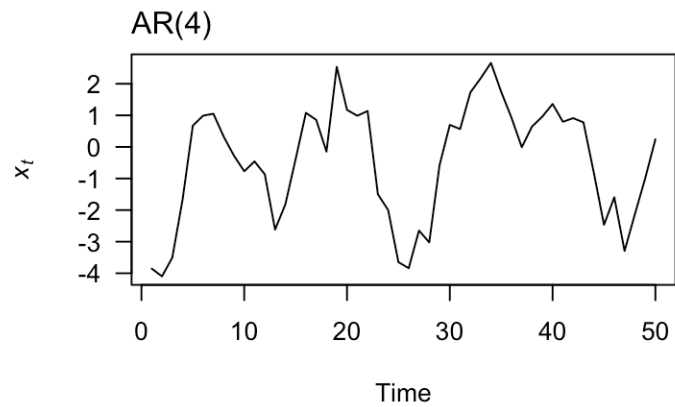
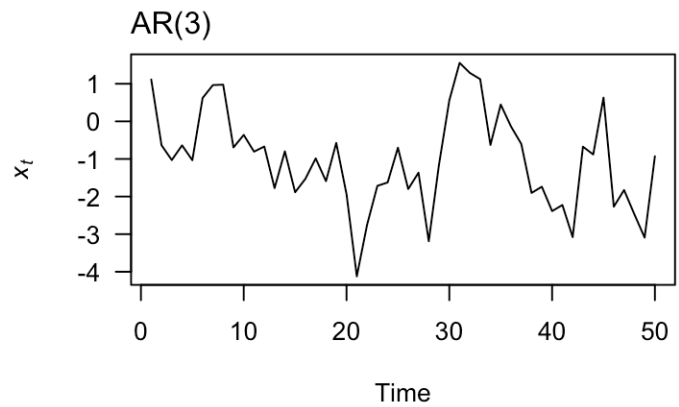
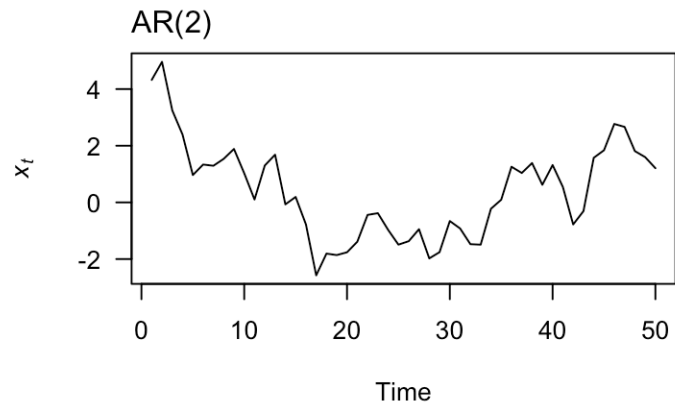
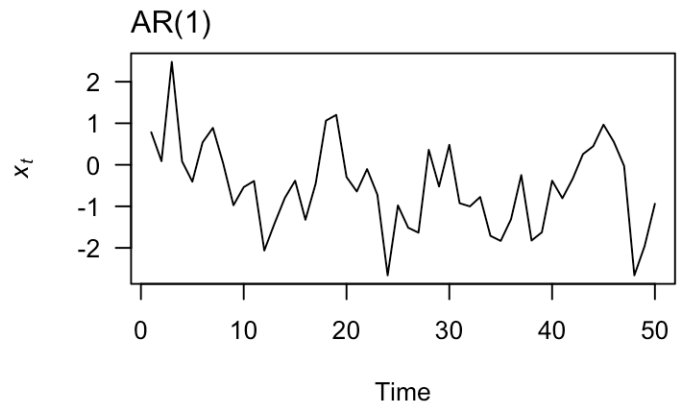
AR(1) with $\phi_1 = 1$ (random walk)

$$x_t = x_{t-1} + w_t$$

AR(2)

$$x_t = -0.2x_{t-1} + 0.4x_{t-2} + w_t$$

Examples of $AR(p)$ models



Stationary AR(p) models

Recall that *stationary* processes have the following properties

1. no systematic change in the mean or variance
2. no systematic trend
3. no periodic variations or seasonality

We seek a means for identifying whether our AR(p) models are also stationary

Stationary AR(p) models

We can write out an AR(p) model using the backshift operator

$$x_t = \phi_1 x_{t-1} + \phi_2 x_{t-2} + \cdots + \phi_p x_{t-p} + w_t$$

$\Downarrow$

$$x_t - \phi_1 x_{t-1} - \phi_2 x_{t-2} - \cdots - \phi_p x_{t-p} = w_t$$

$$(1 - \phi_1 \mathbf{B} - \phi_2 \mathbf{B}^2 - \cdots - \phi_p \mathbf{B}^p) x_t = w_t$$

$$\phi_p (\mathbf{B}^p) x_t = w_t$$

Stationary AR(p) models

If we treat $\mathbf{B}$ as a number (or numbers), we can out write the *characteristic equation* as

$$\phi_p(\mathbf{B})x_t = w_t$$

$\Downarrow$

$$\phi_p(\mathbf{B}^p) = 0$$

To be stationary, **all roots** of the characteristic equation **must exceed 1 in absolute value**

Stationary AR(p) models

For example, consider this AR(1) model from earlier

$$x_t = 0.5x_{t-1} + w_t$$

Stationary AR(p) models

For example, consider this AR(1) model from earlier

$$x_t = 0.5x_{t-1} + w_t$$

$$\Downarrow$$

$$x_t - 0.5x_{t-1} = w_t$$

$$x_t - 0.5\mathbf{B}x_t = w_t$$

$$(1 - 0.5\mathbf{B})x_t = w_t$$

Stationary AR(p) models

For example, consider this AR(1) model from earlier

$$(1 - 0.5\mathbf{B})x_t = w_t$$

$$\Downarrow$$

$$1 - 0.5\mathbf{B} = 0$$

$$-0.5\mathbf{B} = -1$$

$$\mathbf{B} = 2$$

This model is indeed stationary because $\mathbf{B} > 1$

Stationary AR(p) models

What about this AR(2) model from earlier?

$$x_t = -0.2x_{t-1} + 0.4x_{t-2} + w_t$$

Stationary AR(p) models

What about this AR(2) model from earlier?

$$x_t = -0.2x_{t-1} + 0.4x_{t-2} + w_t$$

$\Downarrow$

$$x_t + 0.2x_{t-1} - 0.4x_{t-2} = w_t$$

$$x_t + 0.2\mathbf{B}x_t - 0.4\mathbf{B}^2x_t = w_t$$

$$(1 + 0.2\mathbf{B} - 0.4\mathbf{B}^2)x_t = w_t$$

Stationary AR(p) models

What about this AR(2) model from earlier?

$$(1 + 0.2\mathbf{B} - 0.4\mathbf{B}^2)x_t = w_t$$

$$\Downarrow$$

$$1 + 0.2\mathbf{B} - 0.4\mathbf{B}^2 = 0$$

$$\Downarrow$$

$$\mathbf{B}_1 \approx -1.35 \text{ and } \mathbf{B}_2 \approx 1.85$$

This model is *not* stationary because only $\mathbf{B}_2 > 1$

What about random walks?

Consider our random walk model

$$x_t = x_{t-1} + w_t$$

What about random walks?

Consider our random walk model

$$x_t = x_{t-1} + w_t$$

$$\Downarrow$$

$$x_t - x_{t-1} = w_t$$

$$x_t - \mathbf{1B}x_t = w_t$$

$$(1 - \mathbf{1B})x_t = w_t$$

What about random walks?

Consider our random walk model

$$x_t - x_{t-1} = w_t$$

$$x_t - 1\mathbf{B}x_t = w_t$$

$$(1 - 1\mathbf{B})x_t = w_t$$

$$\Downarrow$$

$$1 - 1\mathbf{B} = 0$$

$$-1\mathbf{B} = -1$$

$$\mathbf{B} = 1$$

Random walks are **not** stationary because $\mathbf{B} = 1 \not\prec 1$

Stationary AR(1) models

We can define a parameter space over which all AR(1) models are stationary

$$x_t = \phi x_{t-1} + w_t$$

Stationary AR(1) models

We can define a parameter space over which all AR(1) models are stationary

$$x_t = \phi x_{t-1} + w_t$$

$$\Downarrow$$

$$x_t - \phi x_{t-1} = w_t$$

$$x_t - \phi \mathbf{B} x_t = w_t$$

$$(1 - \phi \mathbf{B}) x_t = w_t$$

Stationary AR(1) models

For $x_t = \phi x_{t-1} + w_t$, we have

$$(1 - \phi \mathbf{B})x_t = w_t$$

$$\Downarrow$$

$$1 - \phi \mathbf{B} = 0$$

$$-\phi \mathbf{B} = -1$$

$$\mathbf{B} = \frac{1}{\phi}$$

$$\Downarrow$$

$$\mathbf{B} = \frac{1}{\phi} > 1 \text{ iff } 0 < \phi < 1$$

Stationary AR(1) models

What if ϕ is negative, such that $x_t = -\phi x_{t-1} + w_t$?

$$x_t = -\phi x_{t-1} + w_t$$

$$\Downarrow$$

$$x_t + \phi x_{t-1} = w_t$$

$$x_t + \phi \mathbf{B} x_t = w_t$$

$$(1 + \phi \mathbf{B}) x_t = w_t$$

Stationary AR(1) models

For $x_t = -\phi x_{t-1} + w_t$, we have

$$(1 + \phi \mathbf{B})x_t = w_t$$

$$\Downarrow$$

$$1 + \phi \mathbf{B} = 0$$

$$\phi \mathbf{B} = -1$$

$$\mathbf{B} = -\frac{1}{\phi}$$

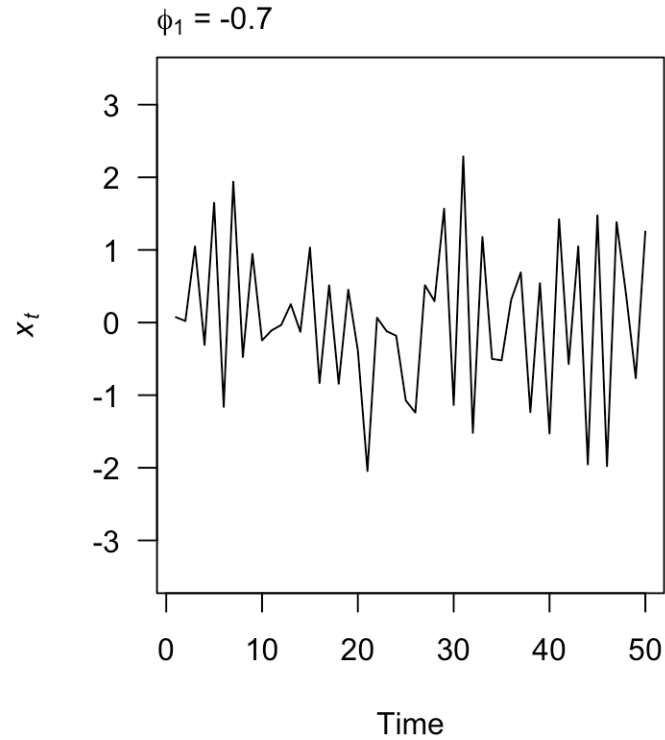
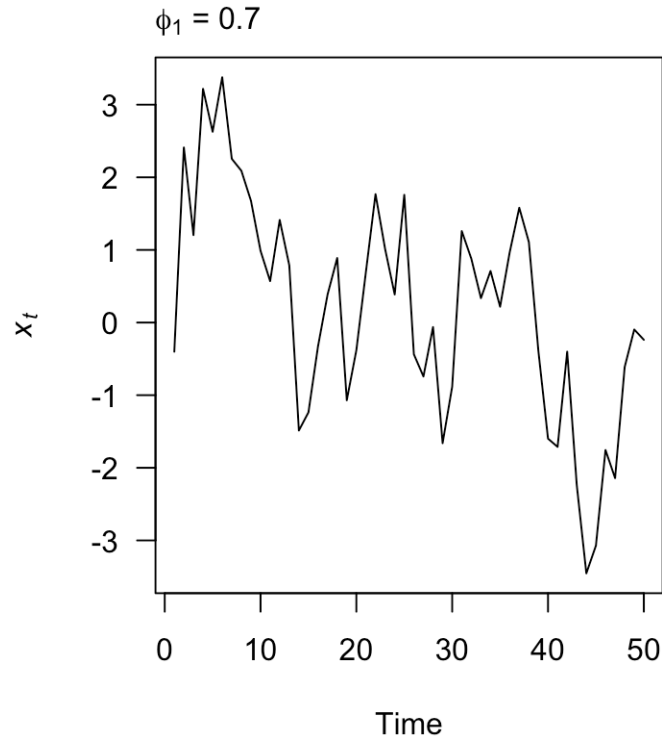
$$\Downarrow$$

$$\mathbf{B} = -\frac{1}{\phi} > 1 \text{ iff } -1 < \phi < 0$$

Stationary AR(1) models

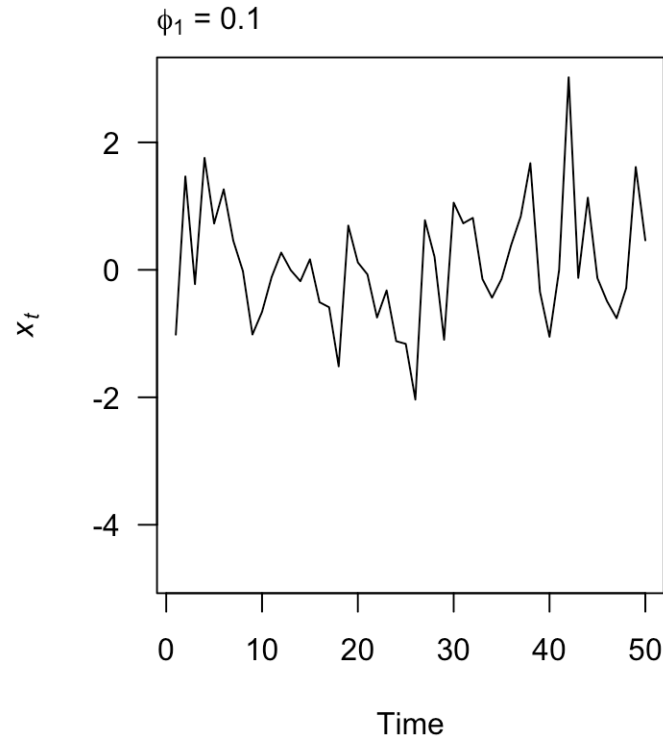
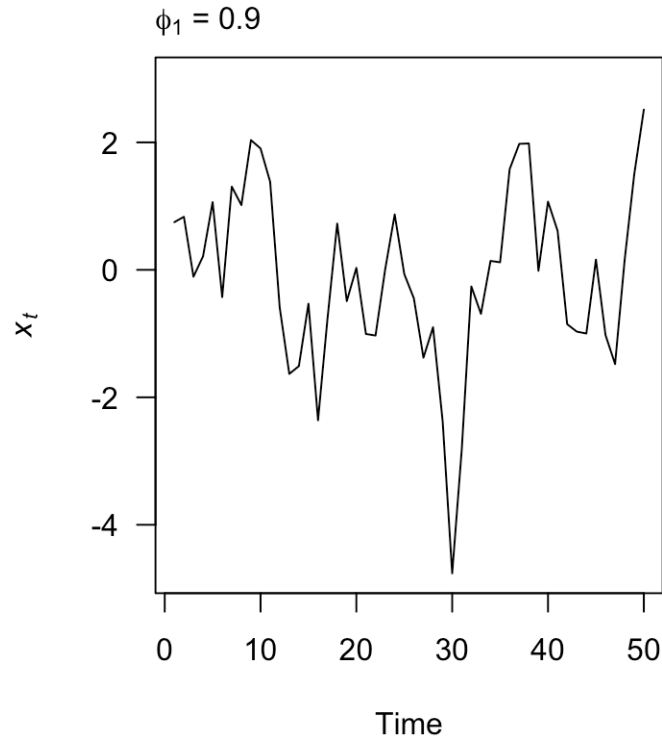
Thus, AR(1) models are stationary if and only if $|\phi| < 1$

Coefficients of AR(1) models



Same value, but different sign

Coefficients of AR(1) models

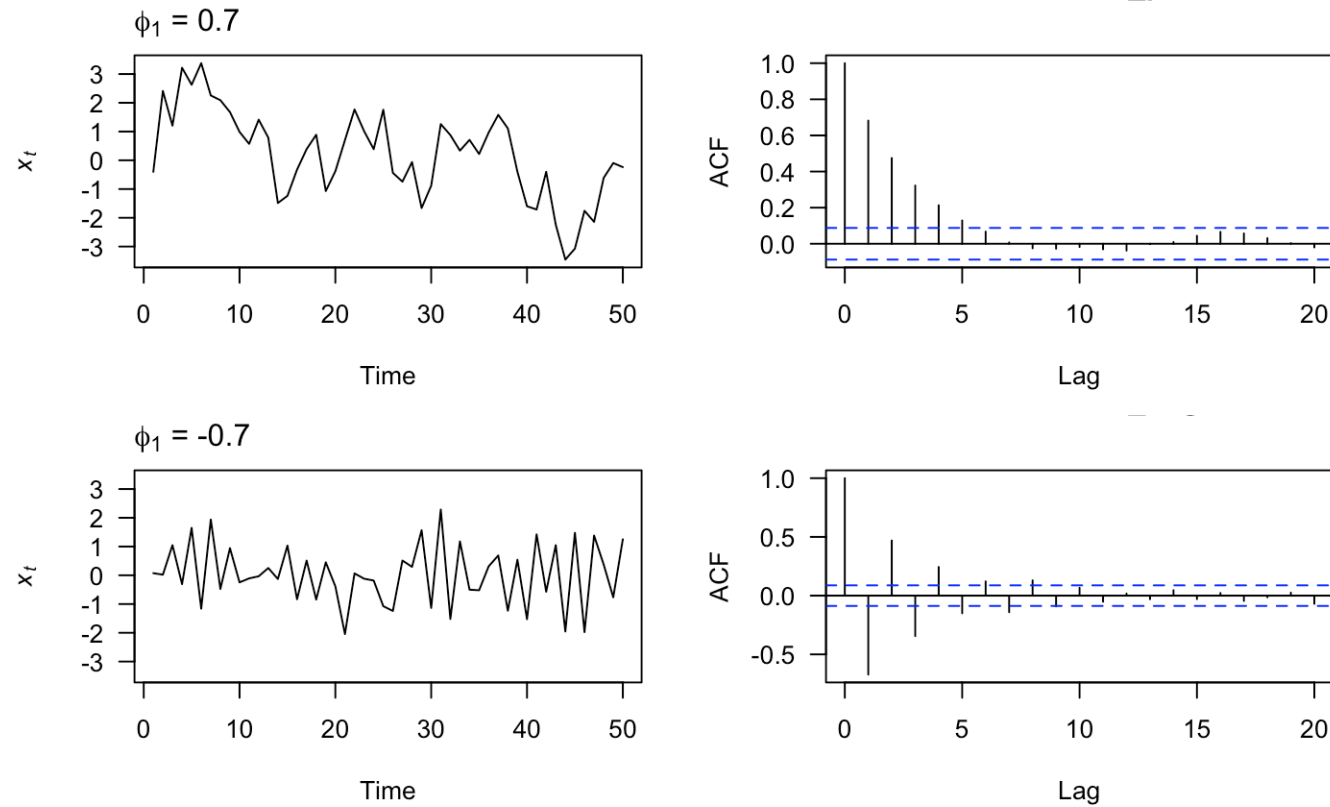


Both positive, but different magnitude

Autocorrelation function (ACF)

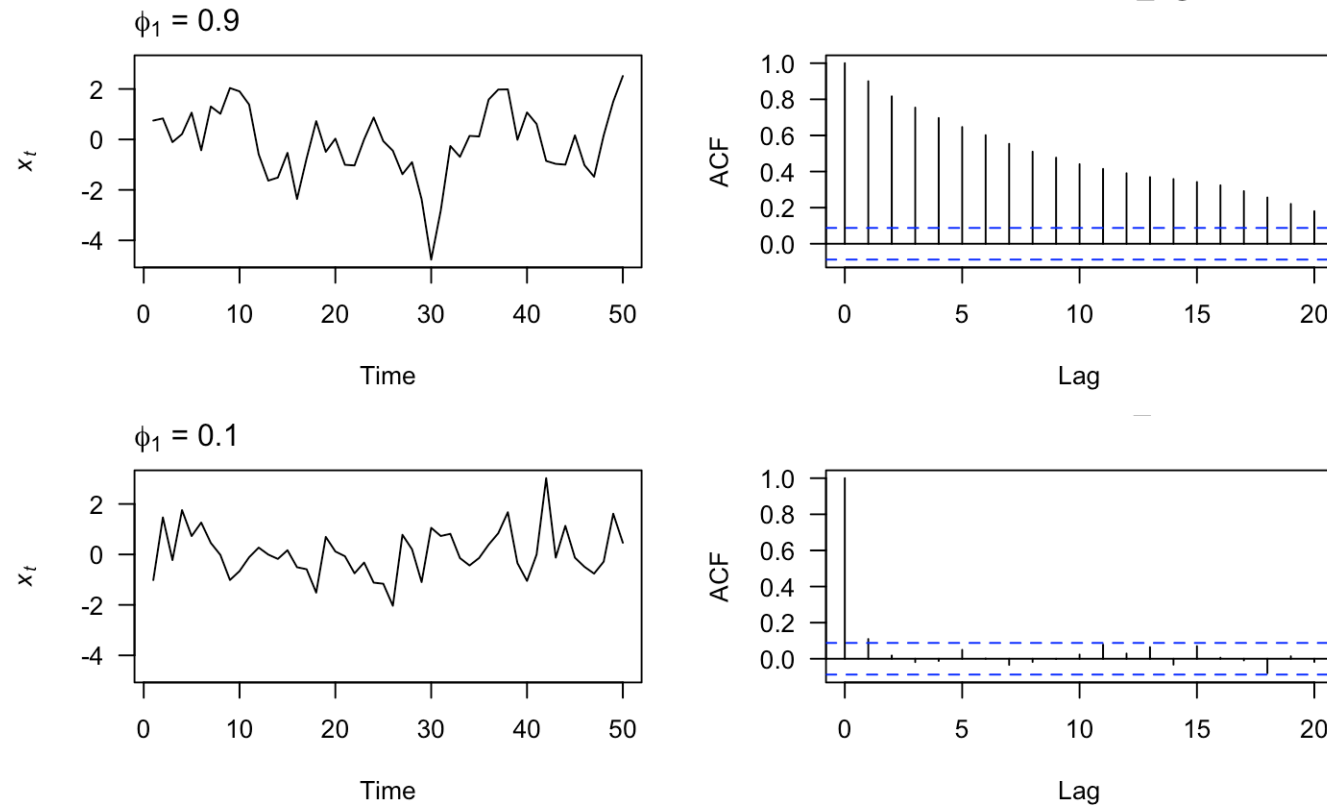
Recall that the *autocorrelation function* (ρ_k) measures the correlation between $\{x_t\}$ and a shifted version of itself $\{x_{t+k}\}$

ACF for AR(1) models



ACF oscillates for model with $-\phi$

ACF for AR(1) models



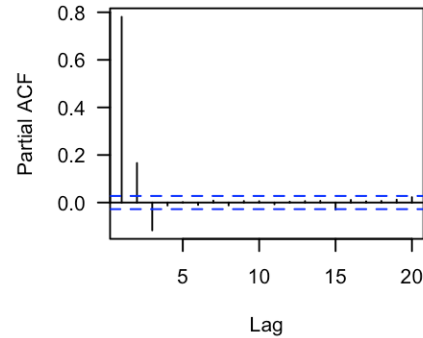
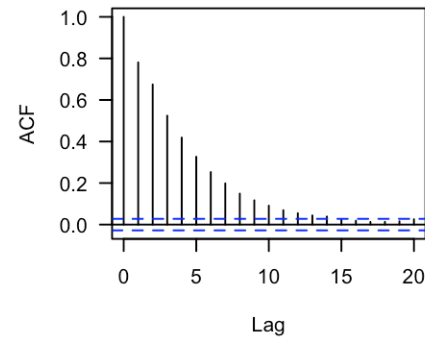
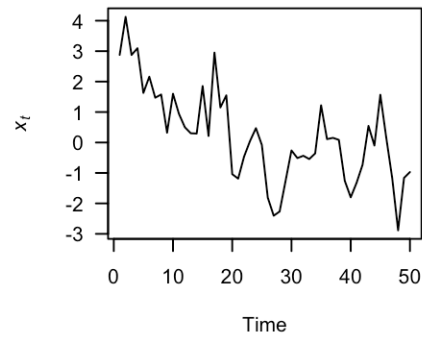
For model with large ϕ , ACF has longer tail

Partial autocorrelation function (PACF)

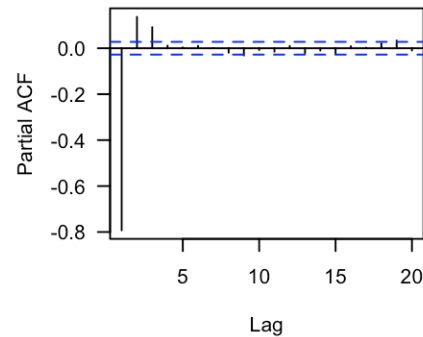
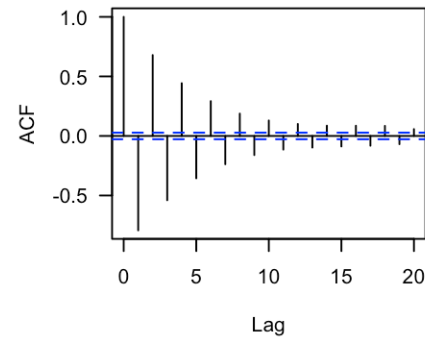
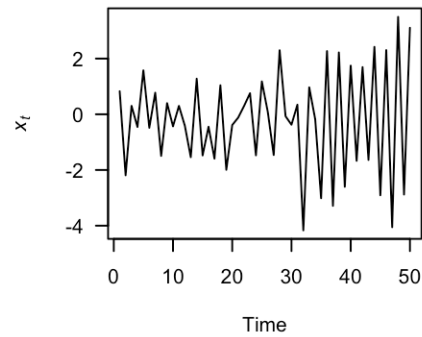
Recall that the *partial autocorrelation function* (ϕ_k) measures the correlation between $\{x_t\}$ and a shifted version of itself $\{x_{t+k}\}$, with the linear dependence of $\{x_{t-1}, x_{t-2}, \dots, x_{t-k-1}\}$ removed

ACF & PACF for AR(p) models

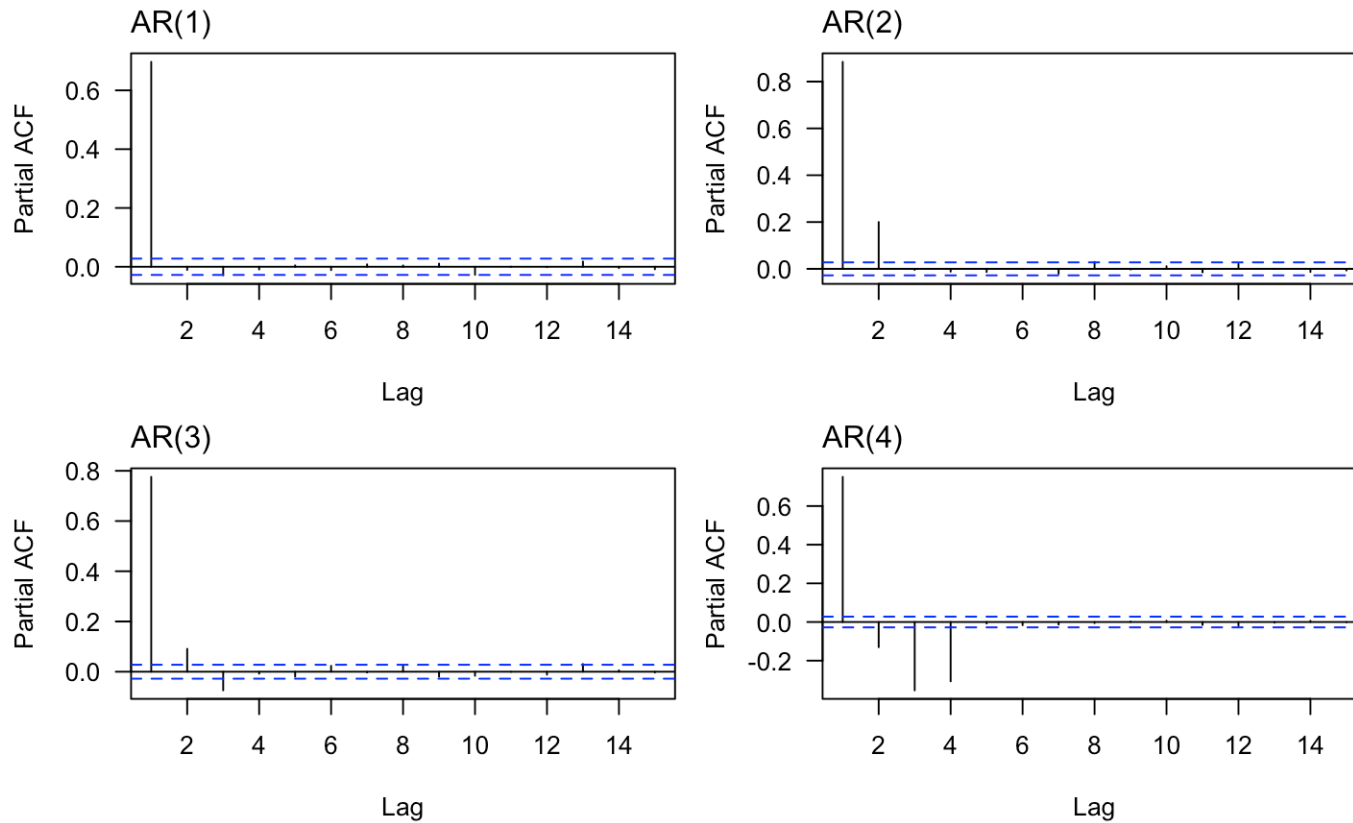
AR(3) with $\phi_1 = 0.7$, $\phi_2 = 0.2$, $\phi_3 = -0.1$



AR(3) with $\phi_1 = -0.7$, $\phi_2 = 0.2$, $\phi_3 = 0.1$



PACF for $AR(p)$ models



Do you see the link between the order p and lag k ?

Using ACF & PACF for model ID

Model

ACF

PACF

$AR(p)$

Tails off slowly

Cuts off after lag p

Moving average (MA) models

Moving average models are most commonly used for forecasting a future state

Moving average (MA) models

A moving average model of order q , or $MA(q)$, is defined as

$$x_t = w_t + \theta_1 w_{t-1} + \theta_2 w_{t-2} + \cdots + \theta_q w_{t-q}$$

where w_t is white noise

Each of the x_t is a sum of the most recent error terms

Moving average (MA) models

A moving average model of order q , or MA(q), is defined as

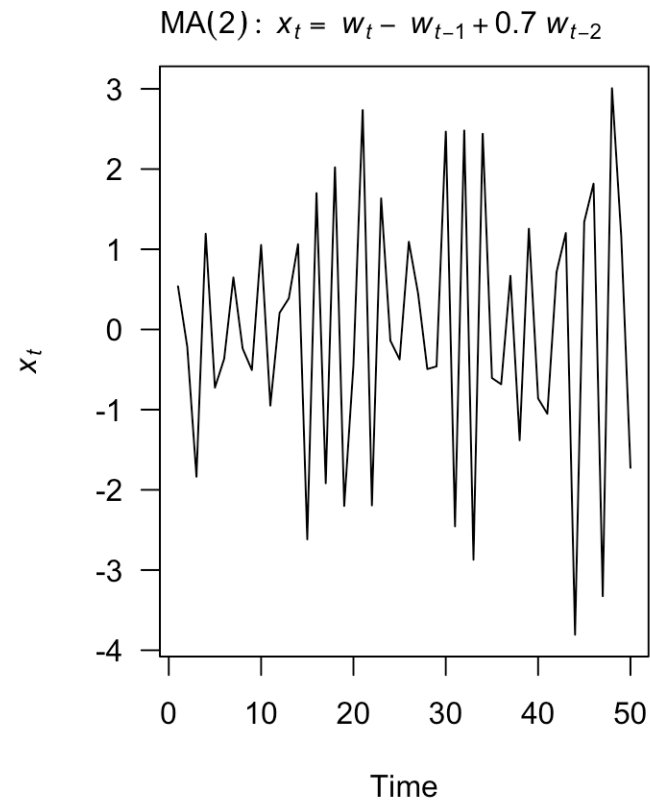
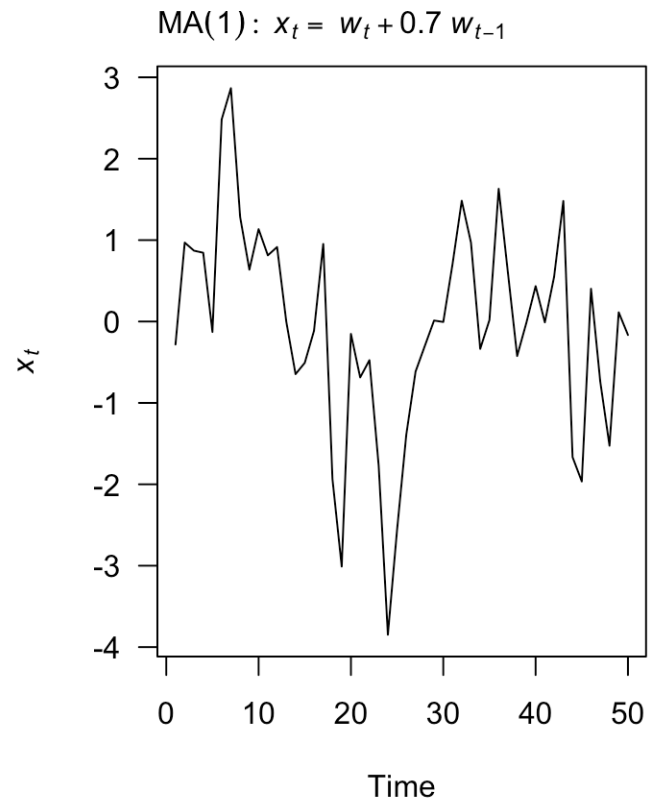
$$x_t = w_t + \theta_1 w_{t-1} + \theta_2 w_{t-2} + \cdots + \theta_q w_{t-q}$$

where w_t is white noise

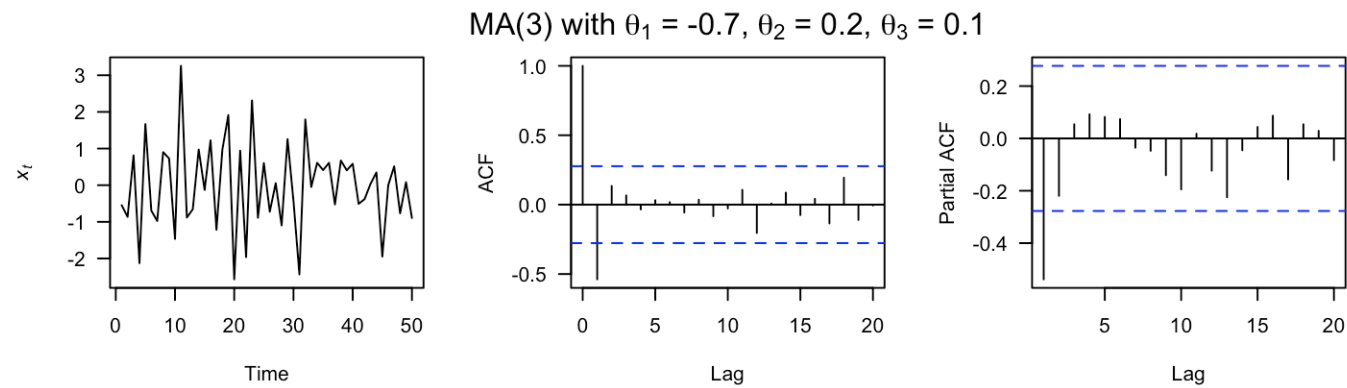
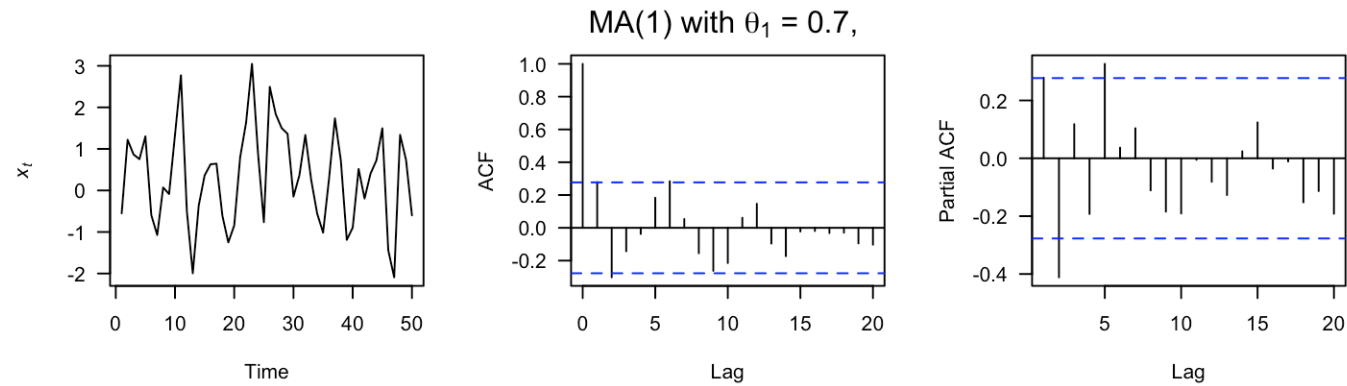
Each of the x_t is a sum of the most recent error terms

Thus, *all* MA processes are stationary because they are finite sums of stationary WN processes

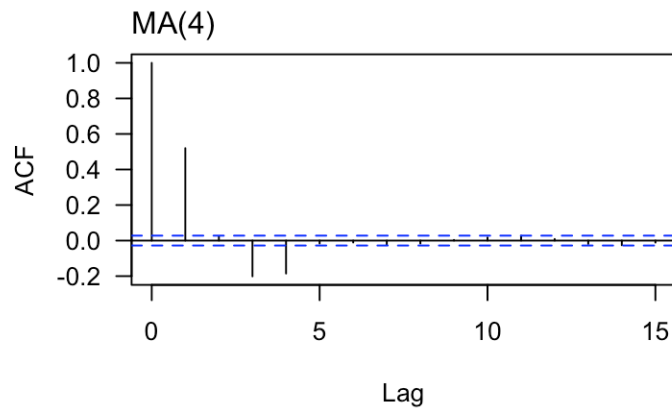
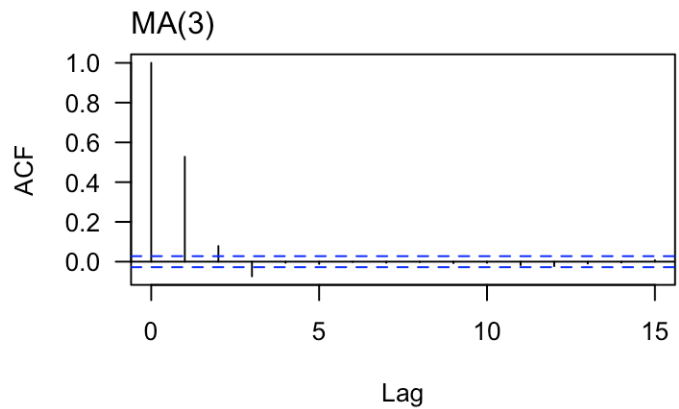
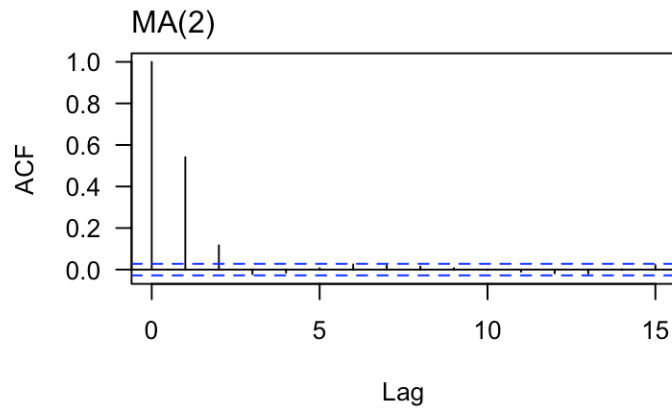
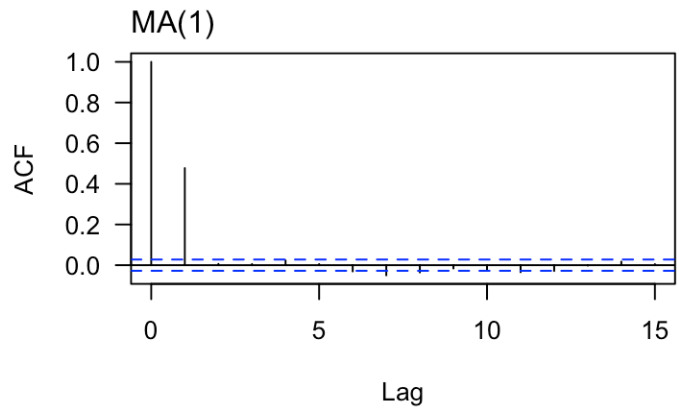
Examples of MA(q) models



ACF & PACF for MA(q) models



ACF for MA(q) models



Do you see the link between the order q and lag k ?

Using ACF & PACF for model ID

Model

ACF

PACF

$AR(p)$

Tails off slowly

Cuts off after lag p

$MA(q)$

Cuts off after lag q

Tails off slowly

AR(p) model as an MA(∞) model

It is possible to write an AR(p) model as an MA(∞) model

AR(1) model as an MA(∞) model

For example, consider an AR(1) model

$$x_t = \phi x_{t-1} + w_t$$

AR(1) model as an MA(∞) model

For example, consider an AR(1) model

$$\begin{aligned}x_t &= \phi x_{t-1} + w_t \\&\Downarrow \\x_{t-1} &= \phi x_{t-2} + w_{t-1} \\&\Downarrow \\x_{t-2} &= \phi x_{t-3} + w_{t-2} \\&\Downarrow \\x_{t-3} &= \phi x_{t-4} + w_{t-3}\end{aligned}$$

AR(1) model as an MA(∞) model

Substituting in the expression for x_{t-1} into that for x_t

$$x_t = \phi x_{t-1} + w_t$$

$$\Downarrow$$

$$x_{t-1} = \phi x_{t-2} + w_{t-1}$$

$$\Downarrow$$

$$x_t = \phi(\phi x_{t-2} + w_{t-1}) + w_t$$

$$x_t = \phi^2 x_{t-2} + \phi w_{t-1} + w_t$$

AR(1) model as an MA(∞) model

And repeated substitutions yields

$$x_t = \phi^2 x_{t-2} + \phi w_{t-1} + w_t$$

$$\Downarrow$$

$$x_t = \phi^3 x_{t-3} + \phi^2 w_{t-2} + \phi w_{t-1} + w_t$$

$$\Downarrow$$

$$x_t = \phi^4 x_{t-4} + \phi^3 w_{t-3} + \phi^2 w_{t-2} + \phi w_{t-1} + w_t$$

$$\Downarrow$$

$$x_t = w_t + \phi w_{t-1} + \phi^2 w_{t-2} + \cdots + \phi^k w_{t-k} + \phi^{k+1} x_{t-k-1}$$

AR(1) model as an MA(∞) model

If our AR(1) model is stationary, then

$$|\phi| < 1$$

which then implies that

$$\lim_{k \rightarrow \infty} \phi^{k+1} = 0$$

AR(1) model as an MA(∞) model

If our AR(1) model is stationary, then

$$|\phi| < 1$$

which then implies that

$$\lim_{k \rightarrow \infty} \phi^{k+1} = 0$$

and hence

$$x_t = w_t + \phi w_{t-1} + \phi^2 w_{t-2} + \cdots + \phi^k w_{t-k} + \phi^{k+1} x_{t-k-1}$$

$\Downarrow$

$$x_t = w_t + \phi w_{t-1} + \phi^2 w_{t-2} + \cdots + \phi^k w_{t-k}$$

Invertible MA(q) models

An MA(q) process is *invertible* if it can be written as a stationary autoregressive process of infinite order without an error term

$$x_t = w_t + \theta_1 w_{t-1} + \theta_2 w_{t-2} + \cdots + \theta_q w_{t-q}$$

$\Downarrow?$

$$w_t = x_t + \sum_{k=1}^{\infty} (-\theta)^k x_{t-k}$$

Invertible $MA(q)$ models

Q: Why do we care if an $MA(q)$ model is invertible?

A: It helps us identify the model's parameters

Invertible MA(q) models

For example, these MA(1) models are equivalent

$$x_t = w_t + \frac{1}{5}w_{t-1} \text{ with } w_t \sim N(0, 25)$$

$\Leftrightarrow$

$$x_t = w_t + 5w_{t-1} \text{ with } w_t \sim N(0, 1)$$

Variance of an MA(1) model

The variance of x_t is given by

$$x_t = w_t + \frac{1}{5}w_{t-1} \text{ with } w_t \sim N(0, 25)$$

$\Downarrow$

$$\begin{aligned}\text{Var}(x_t) &= \text{Var}(w_t) + \left(\frac{1}{25}\right) \text{Var}(w_{t-1}) \\ &= 25 + \left(\frac{1}{25}\right) 25 \\ &= 25 + 1 \\ &= 26\end{aligned}$$

Variance of an MA(1) model

The variance of x_t is given by

$$x_t = w_t + 5w_{t-1} \text{ with } w_t \sim N(0, 1)$$

$\Downarrow$

$$\begin{aligned}\text{Var}(x_t) &= \text{Var}(w_t) + (25)\text{Var}(w_{t-1}) \\ &= 1 + (25)1 \\ &= 1 + 25 \\ &= 26\end{aligned}$$

Rewriting an MA(1) model

We can rewrite an MA(1) model in terms of x

$$x_t = w_t + \theta w_{t-1}$$

$$\Downarrow$$

$$w_t = x_t - \theta w_{t-1}$$

Rewriting an MA(1) model

And now we can substitute in previous expressions for w_t

$$\begin{aligned}w_t &= x_t - \theta w_{t-1} \\&\Downarrow \\w_{t-1} &= x_{t-1} - \theta w_{t-2} \\&\Downarrow \\w_t &= x_t - \theta(x_{t-1} - \theta w_{t-2}) \\w_t &= x_t - \theta x_{t-1} - \theta^2 w_{t-2} \\&\vdots \\w_t &= x_t - \theta x_{t-1} - \cdots - \theta^k x_{t-k} - \theta^{k+1} w_{t-k-1}\end{aligned}$$

Invertible MA(1) model

If we constrain $|\theta| < 1$, then

$$\lim_{k \rightarrow \infty} (-\theta)^{k+1} w_{t-k-1} = 0$$

and

$$w_t = x_t - \theta x_{t-1} - \cdots - \theta^k x_{t-k} - \theta^{k+1} w_{t-k-1}$$

$$\Downarrow$$

$$w_t = x_t - \theta x_{t-1} - \cdots - \theta^k x_{t-k}$$

$$w_t = x_t + \sum_{k=1}^{\infty} (-\theta)^k x_{t-k}$$

Autoregressive moving average models

An autoregressive moving average, or ARMA(p, q), model is written as

$$x_t = \phi_1 x_{t-1} + \cdots + \phi_p x_{t-p} + w_t + \theta_1 w_{t-1} + \cdots + \theta_q w_{t-q}$$

Autoregressive moving average models

We can write an ARMA(p, q) model using the backshift operator

$$\phi_p(\mathbf{B}^p)x_t = \theta_q(\mathbf{B}^q)w_t$$

Autoregressive moving average models

We can write an ARMA(p, q) model using the backshift operator

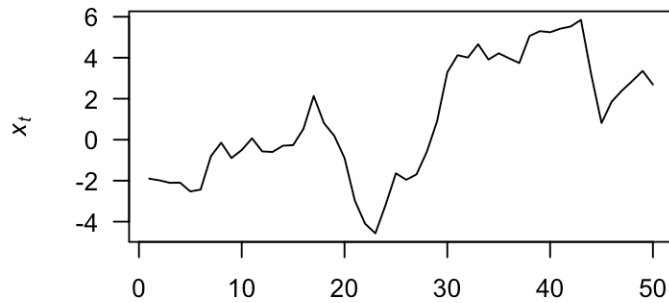
$$\phi_p(\mathbf{B}^p)x_t = \theta_q(\mathbf{B}^q)w_t$$

ARMA models are *stationary* if all roots of $\phi_p(\mathbf{B}) > 1$

ARMA models are *invertible* if all roots of $\theta_q(\mathbf{B}) > 1$

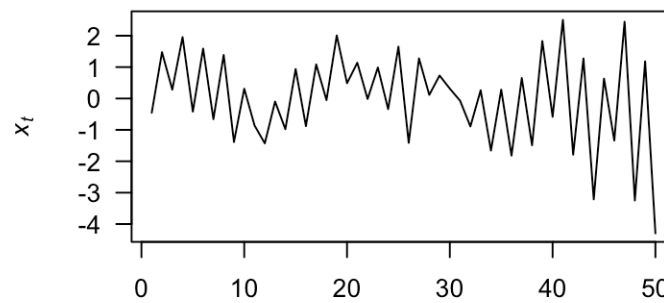
Examples of ARMA(p,q) models

ARMA(3,1): $\phi_1 = 0.7, \phi_2 = 0.2, \phi_3 = -0.1, \theta_1 = 0.5$



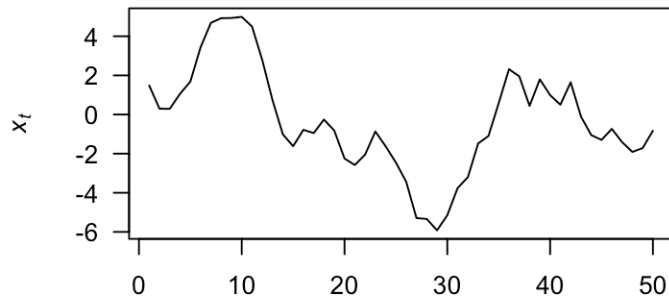
Time

ARMA(2,2): $\phi_1 = -0.7, \phi_2 = 0.2, \theta_1 = 0.7, \theta_2 = 0.2$



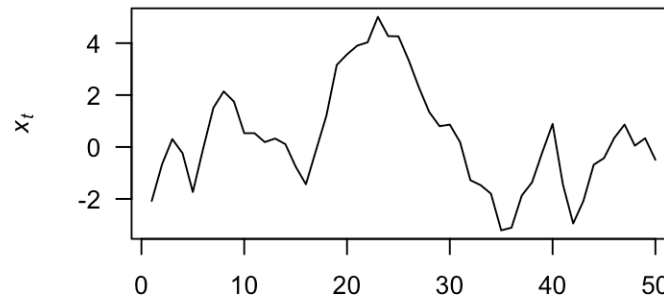
Time

ARMA(1,3): $\phi_1 = 0.7, \theta_1 = 0.7, \theta_2 = 0.2, \theta_3 = 0.5$



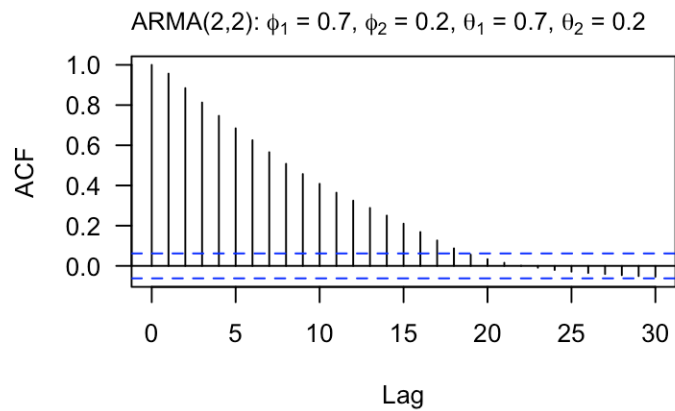
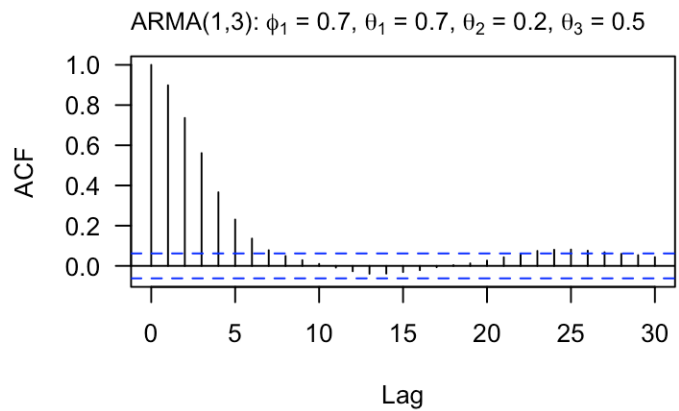
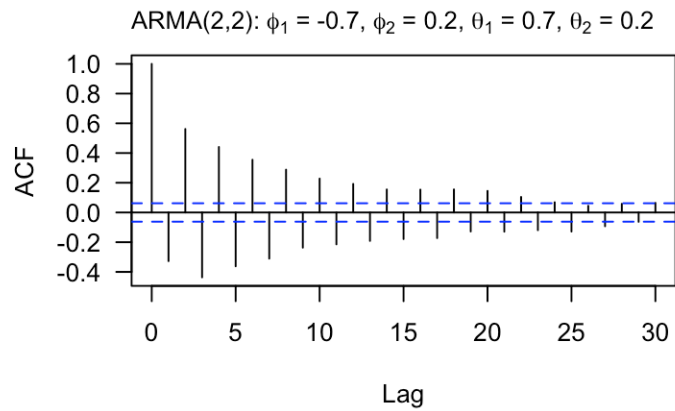
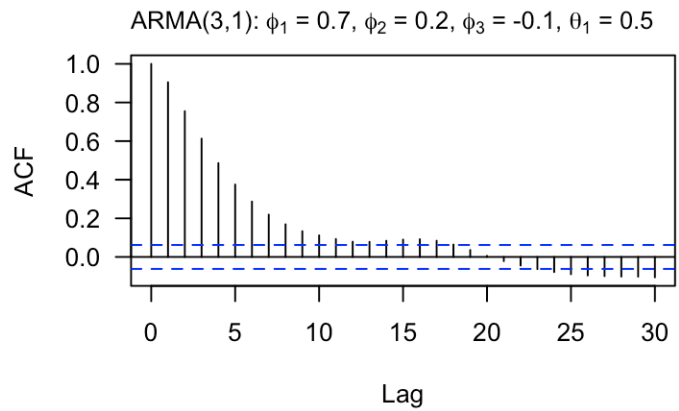
Time

ARMA(2,2): $\phi_1 = 0.7, \phi_2 = 0.2, \theta_1 = 0.7, \theta_2 = 0.2$

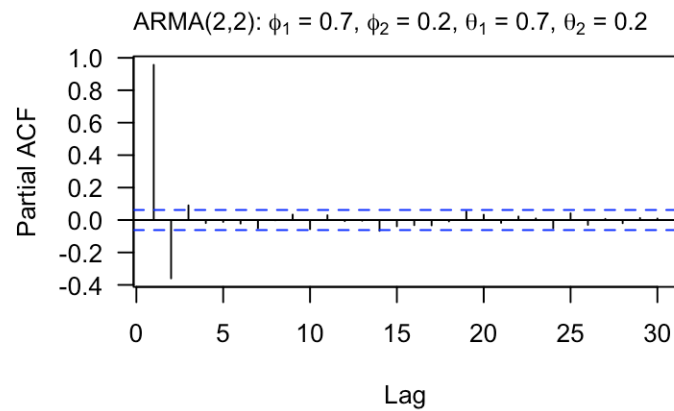
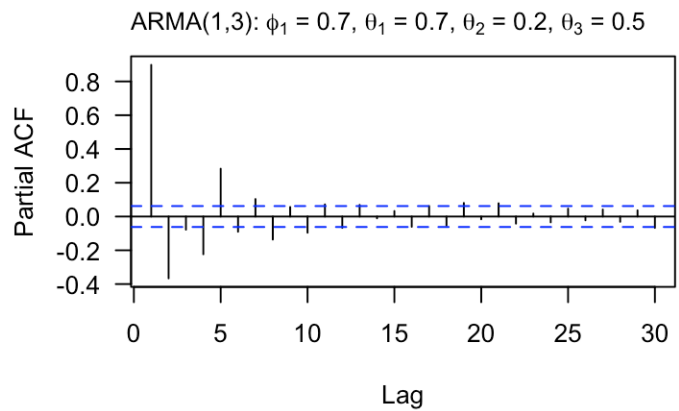
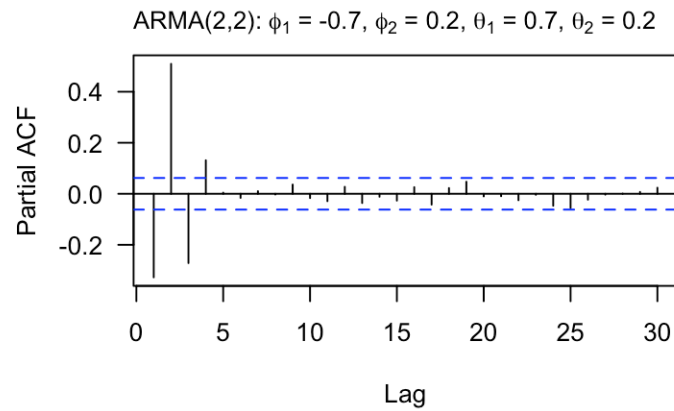
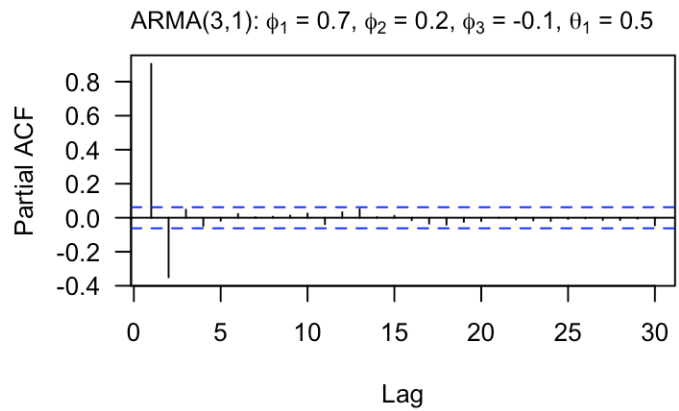


Time

ACF for ARMA(p,q) models



PACF for ARMA(p,q) models



Using ACF & PACF for model ID

Model	ACF	PACF
$AR(p)$	Tails off slowly	Cuts off after lag p
$MA(q)$	Cuts off after lag q	Tails off slowly
$ARMA(p,q)$	Tails off slowly	Tails off slowly

NONSTATIONARY MODELS

Autoregressive integrated moving average (ARIMA) models

If the data do not appear stationary, differencing can help

This leads to the class of *autoregressive integrated moving average* (ARIMA) models

ARIMA models are indexed with orders (p,d,q) where d indicates the order of differencing

ARIMA(p,d,q) models

Definition

$\{x_t\}$ follows an ARIMA(p,d,q) process if $(1 - \mathbf{B})^d x_t$ is an ARMA(p,q) process

ARIMA(p,d,q) models

An example

Consider an ARMA(1,0) = AR(1) process where

$$x_t = (1 + \phi)x_{t-1} + w_t$$

ARIMA(p,d,q) models

An example

Consider an ARMA(1,0) = AR(1) process where

$$x_t = (1 + \phi)x_{t-1} + w_t$$

$$\Downarrow$$

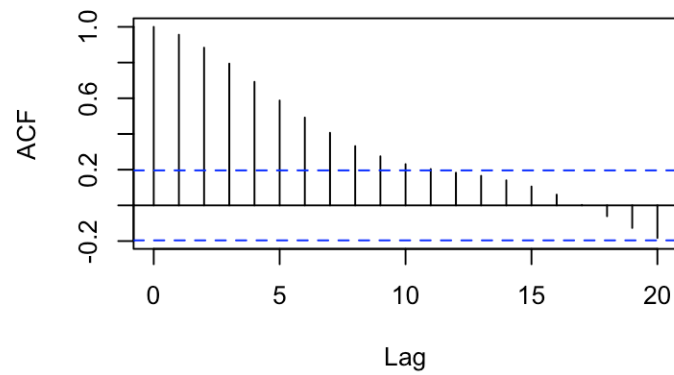
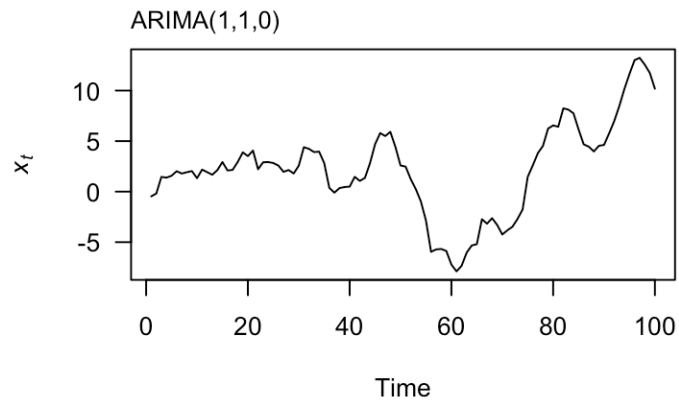
$$x_t = x_{t-1} + \phi x_{t-1} + w_t$$

$$x_t - x_{t-1} = \phi x_{t-1} + w_t$$

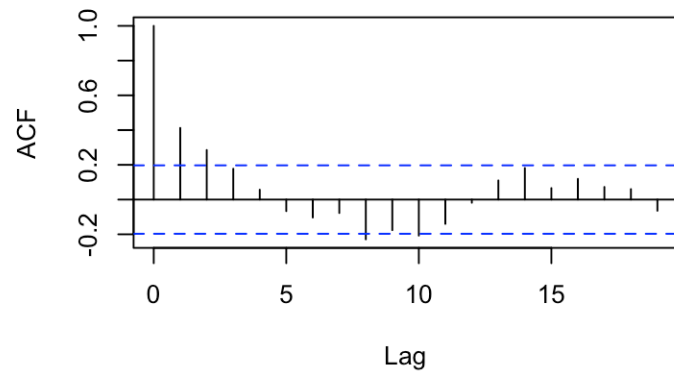
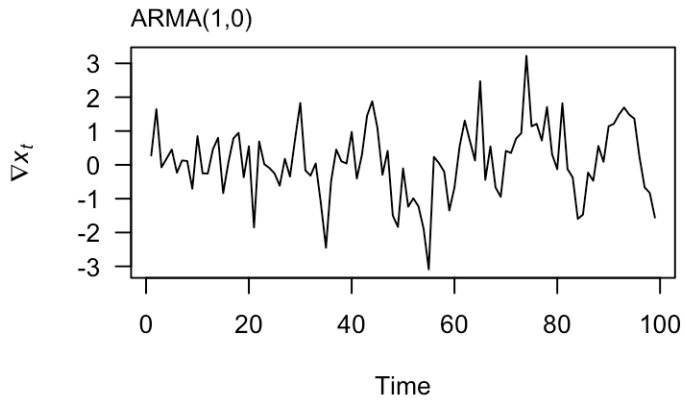
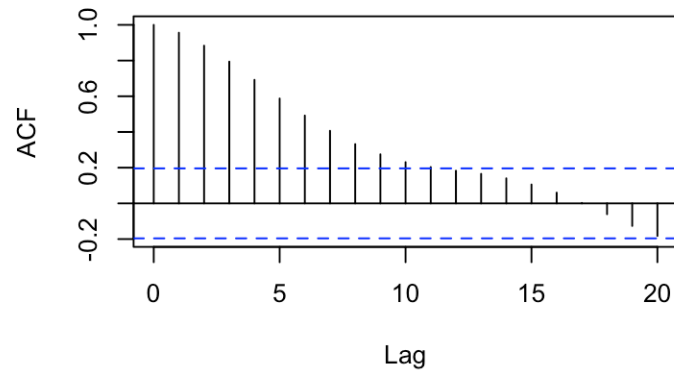
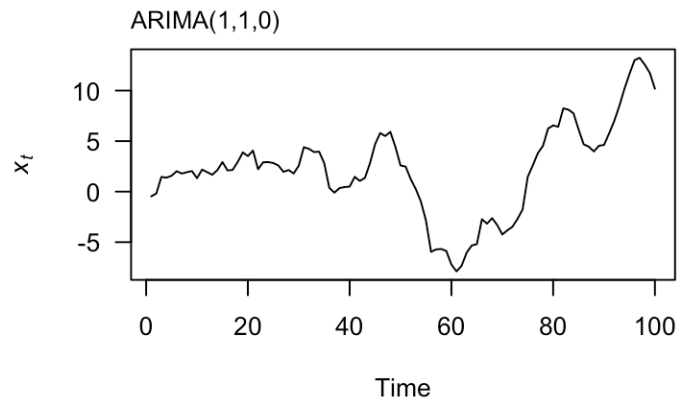
$$(1 - \mathbf{B})x_t = \phi x_{t-1} + w_t$$

So x_t is indeed an ARIMA(1,1,0) process

ARIMA(p,d,q) models



ARIMA(p,d,q) models



Topics for today

Review

- White noise
- Random walks

Autoregressive (AR) models

Moving average (MA) models

Autoregressive moving average (ARMA) models

Using ACF & PACF for model ID